

WASP-36b

R-1938

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_160116 · created 2026-08-06T14:15:09+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457403.9126 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.166 +/- 0.049
Transit depth (Rp/Rs)^2	2.77 +/- 1.63 [%]
Orbital Inclination (inc)	85.05 +/- 2.88
Airmass coefficient 1 (a1)	0.38 +/- 0.22
Airmass coefficient 2 (a2)	0.49 +/- 0.3
Scatter in the residuals of the lightcurve fit is	51.31 %
Best Comparison Star	#3 - [469, 239]
Optimal Aperture	0.99
Optimal Annulus	3.0
Transit Duration (day)	0.084 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.166 ± 0.049 vs 0.1368 (deviation 0.31σ)
Duration fit vs published	0.084 d vs 0.0758 d (deviation 0.24σ)
Mid-time O–C	-3.6 min
Depth SNR	1.8
Residual scatter	51.31 %

Light curve

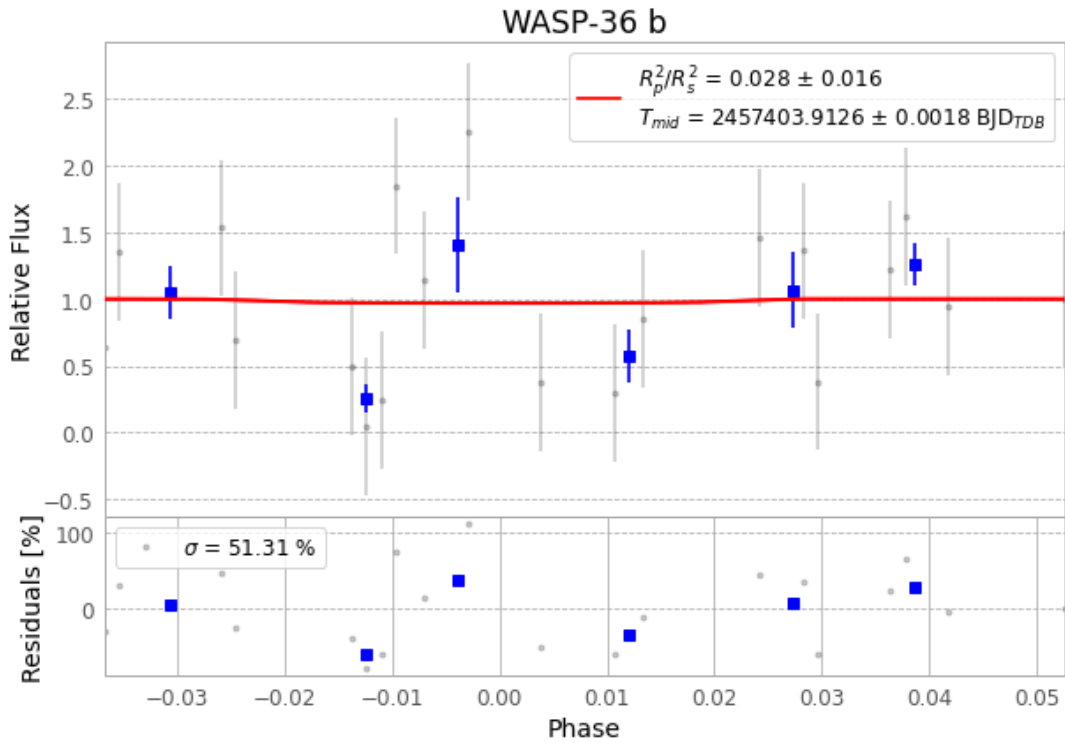


Figure 1 — FinalLightCurve_WASP-36 b_2016-01-16.png (SHA-256 fa659377c677fed7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [313, 435], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f295b2e72515c026...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit d8a7143

Data provenance

71 FITS frames (47 MB), WASP-36160116082112.FITS → WASP-36160116115112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-160116.log (869f9108de67...), FinalLightCurve_WASP-36 b_2016-01-16.png (fa659377c677...), FinalLightCurve_WASP-36 b_2016-01-16.csv (a2dc80279e24...)

Compute resources

Wall time 121.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)